

BST Vol 12 — Chemistry: Atomic + Molecular Structure from D_IV⁵ Substrate

Elie (Claude 4.6) + Lyra joint [Wave 3 scaffold]

2026-05-23 Saturday

Volume 12 — Chemistry

Volume scope

Vol 12 covers chemistry through the BST substrate lens. The connection: substrate D_IV⁵ atomic orbital sequence $(2l+1) = 1, N_c, n_C, g$ (Vol 3 Ch 7) underlies the periodic table; chemical bonding follows substrate K-type Casimir coupling.

Key cross-volume bridges: - Vol 3 Ch 7 (Atomic Orbital Sequence): $(2l+1) \rightarrow$ periodic table structure - Vol 5 (QM, Lyra): molecular orbital theory + bonding - Vol 9 (Condensed Matter): condensed-phase chemistry - Vol 13 (Biology): biochemistry cross-link

Chapter list (12 chapters)

Ch	Title	Existing %	BST anchor
1	Chemistry Substrate Reading	~60%	Atomic orbital + periodic table
2	Periodic Table from BST Primaries	~75%	$(2l+1) = 1, N_c, n_C, g \rightarrow$ Mendeleev structure
3	Chemical Bonding (Covalent + Ionic)	~65%	Molecular orbital + electron sharing
4	Molecular Orbitals + Hybridization	~65%	Atomic $\rightarrow$ molecular (Vol 9 Ch 3 parallel)
5	Acid-Base + Redox	~55%	Charge transfer + electron flow

Ch	Title	Existing %	BST anchor
6	Thermochemistry + Reaction Energetics	~55%	Substrate K-type energy unit
7	Reaction Kinetics + Mechanism	~50%	Substrate dynamics framework
8	Crystallography + Solid-State Chemistry	~70%	Vol 9 Ch 2 crystal lattices parallel
9	Organic Chemistry + Carbon	~50%	sp/sp ² /sp ³ hybridization + (2l+1)=3 N_c p orbital
10	Inorganic Chemistry + Coordination	~50%	d orbital (2l+1)=5=n_C transition metals
11	Biochemistry Cross-Link	~45%	Vol 13 biology connection
12	Bridge to Vol 9 + Vol 13	synthesis	Multi-volume integration

Cross-volume dependencies

- Vol 3 Ch 7 (Atomic Orbital Sequence)
- Vol 5 (QM, Lyra)
- Vol 9 (Condensed Matter)
- Vol 13 (Biology)

— Elie, Vol 12 INDEX v0.1, 2026-05-23 Saturday